# REPRODUCTION IN ORGANISMS

*Objective Type Questions — Bihar Board Class 12 Biology*

Total Questions: 100 • Full Marks: 100 • Time: 2 Hours • Each Question: 1 Mark

**1. Which of the following is unisexual?** [1]

(A) Papaya (B) Mustard

(C) Hibiscus (D) None of these

**2. Statement-I: Male gametes are haploid.**
**Statement-II: Male gametes contain $2n$ chromosomes.** [1]

(A) Both statements are correct and Statement-II is the correct explanation of Statement-I.

(B) Statement-I is correct, but Statement-II is incorrect.

(C) Both statements are correct, but Statement-II is not the correct explanation of Statement-I.

(D) Statement-I is incorrect, but Statement-II is correct.

**3. What conclusion can be drawn by comparing the body size and lifespan of a crow and a parrot?** [1]

(A) Lifespan is directly proportional to body size. (B) Lifespan is inversely proportional to body size.

(C) All organisms have the same lifespan. (D) Lifespan does not depend on body size.

**4. Which has a longer lifespan, peepal or mango?** [1]

(A) Mango (B) Both have the same lifespan.

(C) Peepal (D) Cannot be compared

**5. Why is the exception of natural death observed in unicellular organisms?** [1]

(A) They do not reproduce.

(B) During division, the body of the parent cell is divided into daughter cells.

(C) They never undergo cell division.

(D) They are only very small.

**6. What are clones?** [1]

(A) Genetically different offspring (B) Sexually produced offspring

(C) Genetically identical offspring (D) None of these

**7. Reproduction is essential for organisms because—** [1]

(A) it ensures continuity of the species. (B) it helps in growth.

(C) it provides energy. (D) it provides nutrition.

**8. Approximately how long can the adult stage of some butterflies live?** [1]

(A) 1–2 days (B) 1–2 months

(C) 1–2 years (D) 1–2 weeks

**9. The production of gametes is called:** [1]

(A) Cytokinesis (B) Gametogenesis

(C) Sporogenesis (D) Meiocyte

**10. When offspring are produced by a single parent, it is called:** [1]

(A) Sexual reproduction (B) Both types of reproduction

(C) Asexual reproduction (D) Internal fertilisation

**11. In which of the following does reproduction by budding occur?** [1]

(A) Paramecium (B) Penicillium

(C) All of these (D) Yeast

**12. Which of the following is oviparous?** [1]

(A) Hen (B) Frog

(C) Crocodile (D) All of these

**13. Which of the following is diploid?** [1]

(A) Egg (B) Zygote

(C) Pollen (D) Both egg and pollen

**14. What does Amoeba do under unfavourable conditions?** [1]

(A) Dies (B) Divides

(C) Buds (D) Undergoes encystation

**15. What type of division occurs in encysted Amoeba?** [1]

(A) Multiple fission (B) Binary fission

(C) Budding (D) Fragmentation

**16. What is sporulation?** [1]

(A) Budding (B) Fragmentation

(C) Spore formation (D) Binary fission

**17. Which asexual reproductive structure occurs in Chlamydomonas?** [1]

(A) Zoospore (B) Conidia

(C) Bud (D) Gemmule

**18. Which asexual reproductive structure occurs in Penicillium?** [1]

(A) Zoospore (B) Bud

(C) Conidia (D) Gemmule

**19. How does asexual reproduction occur in Hydra?** [1]

(A) Binary fission (B) Fragmentation

(C) Spore formation (D) Budding

**20. What is vegetative propagation?** [1]

(A) Sexual reproduction (B) Asexual reproduction

(C) Both (D) None of these

**21. From which part does vegetative propagation occur in potato?** [1]

(A) Eyes (B) Rhizome

(C) Bulb (D) Runner

**22. From which part does vegetative propagation occur in ginger?** [1]

(A) Eyes (B) Bulb

(C) Runner (D) Rhizome

**23. From which part does vegetative propagation occur in Bryophyllum?** [1]

(A) Root (B) Leaf

(C) Stem (D) Flower

**24. From which part does vegetative propagation occur in water hyacinth?** [1]

(A) Rhizome (B) Bulb

(C) Runner (D) Bulbil

**25. Vegetative propagation in Agave occurs through:** [1]

(A) Rhizome (B) Bulb

(C) Runner (D) Bulbil

**26. In which does asexual reproduction by fragmentation occur?** [1]

(A) Spirogyra (B) Yeast

(C) Hydra (D) Penicillium

**27. In most eukaryotic organisms, what type of cell division generally occurs during asexual reproduction?** [1]

(A) Meiosis (B) Mitosis

(C) Both (D) None of these

**28. Which of the following is not a method of asexual reproduction?** [1]

(A) Budding (B) Fragmentation

(C) Fertilisation (D) Spore formation

**29. What is fertilisation?** [1]

(A) Fusion of the egg and secondary nucleus (B) Fusion of male and female gametes

(C) Fusion of the egg and synergid (D) Only division of the zygote

**30. What are isogametes?** [1]

(A) Gametes that look different (B) Male gametes

(C) Gametes that look alike (D) Female gametes

**31. What are heterogametes?** [1]

(A) Gametes that look alike (B) Male gametes

(C) Female gametes (D) Gametes that look different

**32. What is the male gamete called?** [1]

(A) Sperm (B) Egg

(C) Zygote (D) Spore

**33. What is the female gamete called?** [1]

(A) Sperm (B) Zygote

(C) Egg (D) Spore

**34. What is a characteristic feature of a monoecious plant?** [1]

(A) Male and female flowers/reproductive structures occur on the same plant.

(B) Male and female flowers occur on different plants.

(C) Only male flowers occur on the plant.

(D) Only female flowers occur on the plant.

**35. What is a characteristic feature of a dioecious (unisexual) plant?** [1]

(A) Male and female flowers/reproductive structures occur on different plants.

(B) Male and female flowers occur on the same plant.

(C) Only male flowers occur on the plant.

(D) Only female flowers occur on the plant.

**36. What is a characteristic feature of a homothallic organism/thallus?** [1]

(A) Two different thalli are required for sexual reproduction.

(B) Only male sexual structures are present.

(C) Only female sexual structures are present.

(D) Compatible sexual types occur in the same thallus/organism.

**37. What is a characteristic feature of a heterothallic organism/thallus?** [1]

(A) Both compatible sexual types occur in the same thallus.

(B) Different compatible thalli/mating types are required for sexual reproduction.

(C) Only male sexual structures are present.

(D) Only female sexual structures are present.

**38. What type of flowers occur in Cucurbita?** [1]

(A) Unisexual (B) Bisexual

(C) Neuter (D) None of these

**39. Which of the following is a dioecious (unisexual) plant?** [1]

(A) Mustard (B) Maize

(C) Papaya (D) Hibiscus

**40. What type of animal is a cockroach?** [1]

(A) Bisexual (B) Hermaphrodite

(C) None of these (D) Unisexual

**41. What is a staminate flower?** [1]

(A) Male flower (B) Female flower

(C) Bisexual flower (D) Neuter flower

**42. What is a pistillate flower?** [1]

(A) Male flower (B) Female flower

(C) Bisexual flower (D) Neuter flower

**43. What are meiocytes?** [1]

(A) Gametes (B) Zygotes

(C) Gamete-producing cells (D) Spores

**44. How many chromosomes are present in a human meiocyte?** [1]

(A) 23 (B) 92

(C) 12 (D) 46

**45. Statement-I: Cycas is a dioecious plant.**
**Statement-II: Its male and female reproductive structures occur on different plants.** [1]

(A) Both statements are correct, but Statement-II is not the correct explanation of Statement-I.

(B) Both statements are correct and Statement-II is the correct explanation of Statement-I.

(C) Statement-I is correct, but Statement-II is incorrect.

(D) Statement-I is incorrect, but Statement-II is correct.

**46. Which of the following is a diploid structure?** [1]

(A) Ovum (B) Sperm

(C) Zygote (D) All of these

**47. Which of the following is a diploid structure?** [1]
(A) Zygote (B) Ovum
(C) Endosperm (D) All of these

**48. Which of the following is a hermaphroditic animal?** [1]
(A) Hen (B) Snake
(C) Both hen and snake (D) Earthworm

**49. Which of the following animals is not hermaphroditic?** [1]
(A) Tapeworm (B) Earthworm
(C) Housefly (D) Leech

**50. Gametes are generally:** [1]
(A) Diploid (B) Polyploid
(C) None of these (D) Haploid

**51. What is syngamy?** [1]
(A) Embryogenesis (B) Fusion of gametes
(C) Gametogenesis (D) Division

**52. In which of the following does external fertilisation occur?** [1]
(A) Frog (B) Human
(C) Cow (D) Dog

**53. In which of the following does internal fertilisation occur?** [1]
(A) Frog (B) Starfish
(C) Sea urchin (D) Human

**54. Parthenogenesis is found in:** [1]
(A) Human (B) Honeybee
(C) Cow (D) Dog

**55. What is a zygote?** [1]
(A) A haploid cell (B) A triploid cell
(C) A diploid cell (D) A tetraploid cell

**56. What is embryogenesis?** [1]
(A) Development of an embryo from a zygote (B) Gamete formation
(C) Fertilisation (D) Division

**57. What are oviparous organisms?** [1]
(A) Those that give birth to young ones (B) Those that lay eggs
(C) Both (D) None of these

**58. What are viviparous organisms?** [1]
(A) Those that lay eggs (B) Both
(C) None of these (D) Those that give birth to young ones

**59. What type of animals are birds?** [1]
(A) Viviparous (B) Both
(C) Oviparous (D) None of these

**60. What is a major disadvantage of external fertilisation?** [1]
(A) Gametes/embryonic stages are more exposed to predators and adverse environmental conditions.
(B) Gametes are not produced at all.
(C) Male and female are not required.
(D) Fertilisation is always impossible.

**61. Which process occurs during embryogenesis?** [1]
(A) Meiosis (B) Mitosis
(C) Both (D) None of these

**62. What is included among the events that occur after fertilisation in plants?** [1]
(A) Development of the embryo from the zygote, followed by development of the seed and fruit
(B) Only gametogenesis
(C) Fertilisation
(D) Pollination

**63. Where is the zygote formed in external fertilisation?** [1]
(A) In the ovary (B) In the testis
(C) In the external medium (water) (D) None of these

**64. Where is the zygote formed in internal fertilisation?** [1]
(A) In the external medium (B) Inside the male body
(C) None of these (D) Inside the female body

**65. Gemmules are characteristic of:** [1]
(A) Sponges (B) Hydra
(C) Yeast (D) Both Hydra and yeast

**66. In which female animal does the oestrous cycle not occur?** [1]
(A) Cow (B) Dog
(C) Human (D) Rabbit

**67. Binary fission is found in:** [1]
(A) Amoeba (B) Paramecium
(C) None of these (D) Both

**68. The process of seed formation without fertilisation in plants is called:** [1]
(A) Parthenogenesis (B) Apomixis
(C) Double fertilisation (D) Polyembryony

**69. The root of which plant can give rise to a new plant?** [1]
(A) Dahlia (B) Gulmohar
(C) Neem (D) Petunia

**70. Reproduction in yeast mainly occurs by:** [1]
(A) Fragmentation (B) Budding
(C) Pollination (D) All of these

**71. Which of the following plants is called the 'Terror of Bengal'?** [1]
(A) Bamboo (B) Lantana
(C) Parthenium (D) Water hyacinth

**72. Which of the following reproduces only once in its lifetime?** [1]
(A) Bird (B) Guava
(C) Bamboo (D) Monkey

**73. Which of the following is not an example of vegetative propagation?** [1]
(A) Potato (B) Ginger
(C) Bryophyllum (D) Pea

**74. Which of the following statements is false?** [1]
(A) Chara is dioecious.
(B) In the male thallus of Marchantia, antheridia are produced on the antheridiophore.
(C) The reproductive cells of diploid organisms form haploid gametes by meiosis.
(D) Chlamydomonas produces asexual motile spores (zoospores).

**75. Marchantia is a dioecious plant and its male sex organ develops on the—** [1]
(A) Tassel (B) Archegoniophore
(C) Antheridiophore (D) Both (B) and (C)

**76. Which developmental stage is called adolescence?** [1]
(A) Only the reproductive period
(B) The developmental stage from childhood towards adulthood
(C) Old age
(D) Only the embryonic stage

**77. The oestrous cycle occurs in:** [1]
(A) Primates (B) Birds
(C) Reptiles (D) Non-primate mammals

**78. The menstrual cycle occurs in:** [1]
(A) Non-primate mammals (B) Primates
(C) Birds (D) Reptiles

**79. According to NCERT, many bamboo species flower once after approximately how many years?** [1]
(A) 50–100 (B) 10–20
(C) 100–200 (D) 5–10

**80. What is the pericarp?** [1]
(A) Seed (B) Embryo
(C) Protective wall of the fruit (D) Zygote

**81. How often does Strobilanthus kunthiana flower?** [1]
(A) 6 years (B) 18 years
(C) 24 years (D) 12 years

**82. What are seasonal breeders?** [1]
(A) Those that reproduce throughout the year (B) Those that reproduce during favourable seasons
(C) Those that never reproduce (D) None of these

**83. What are continuous breeders?** [1]
(A) Those that reproduce during favourable seasons (B) Those that never reproduce
(C) Those that reproduce throughout the year (D) None of these

**84. What does the ovary become after fertilisation?** [1]
(A) Fruit (B) Seed
(C) Embryo (D) Pericarp

**85. What does the ovule become after fertilisation?** [1]
(A) Fruit (B) Embryo
(C) Seed (D) Pericarp

**86. What is the scientific name of water hyacinth?** [1]
(A) Eichhornia crassipes (B) Lantana camara
(C) Parthenium hysterophorus (D) Hydrilla

**87. Which structure is not involved in vegetative propagation?** [1]
(A) Rhizome (B) Seed
(C) Sucker (D) Tuber

**88. Asexual reproduction in yeast mainly occurs by which method?** [1]
(A) Binary fission (B) Multiple fission
(C) Fragmentation (D) Budding

**89. How many chromosomes are present in a human gamete?** [1]
(A) 23 (B) 46
(C) 92 (D) 12

**90. Which type of division occurs during gamete formation in diploid organisms?** [1]
(A) Mitosis (B) Both
(C) None of these (D) Meiosis

**91. Which type of division occurs during gamete formation in haploid organisms?** [1]
(A) Meiosis (B) Both
(C) Mitosis (D) None of these

**92. Which parts of a flower wither and fall off after fertilisation?** [1]
(A) Ovary (B) Sepals, petals and stamens
(C) Stigma (D) None of these

**93. In parthenogenesis, a new organism develops from the:** [1]
(A) Fertilised egg (B) Zygote
(C) Unfertilised female gamete (D) Spore

**94. A seed develops from the:** [1]
(A) Ovary (B) Ovule
(C) Embryo (D) Pericarp

**95. In diploid organisms, the specialised cells that produce gametes are called:** [1]
(A) Meiocytes (B) Gametes
(C) Zygotes (D) Spores

**96. What is observed in organisms that undergo external fertilisation?** [1]
(A) Very little gamete production (B) Internal development
(C) None of these (D) Synchrony between males and females

**97. In vegetative propagation, new plants arise from:** [1]
(A) Seeds (B) Vegetative structures
(C) Gametes (D) Zygotes

**98. Which of the following is a unicellular organism?** [1]
(A) Amoeba (B) Hydra
(C) Sponge (D) Earthworm

**99. Which statement is true about asexual reproduction?** [1]
(A) It involves fusion of gametes. (B) Two parents take part.
(C) It is a slow process. (D) One parent takes part.

**100. Which statement is true about sexual reproduction?** [1]
(A) One parent takes part. (B) The offspring are clones.
(C) It involves fusion of gametes. (D) It is a rapid process.